

Space-Grade Mechanical Fault Detector

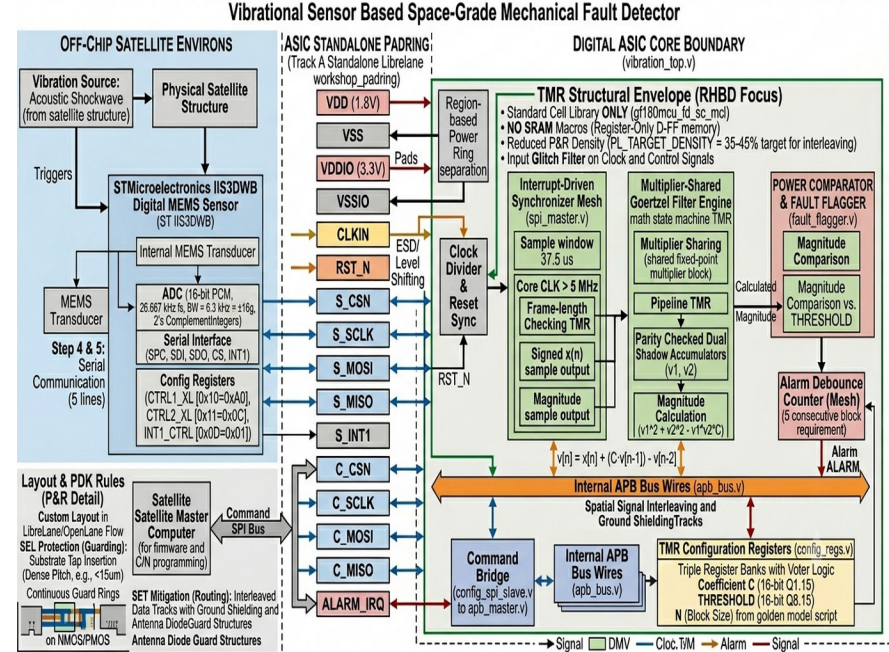
An IEEE SSCS Chipathon 2026 Project — Team B22: Space Jam

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GlobalFoundries 180nm • GF180MCU Process • 600×600 μm Die

System Architecture Overview

- Direct Digital MEMS Interface: 16-bit acceleration samples at 26.667 kHz natively via SPI.
- SPI-APB Interface: Buffers samples & decouples SPI and APB timing domains.
- TMR Register Bank: Radiation-hardened configuration storage for coefficients & thresholds.
- Axis Sequencer: Time-multiplexes X, Y, Z axis data to process them sequentially.
- Goertzel Core: Custom hardware DSP engine for targeted frequency bin filtering.
- Magnitude Compute: Converts recursive filter states into spectral energy snapshot.
- Fault Flagger: Compares spectral magnitude to thresholds and asserts interrupt.
- APB Master: Interfaces with system bus for register reads and configuration.



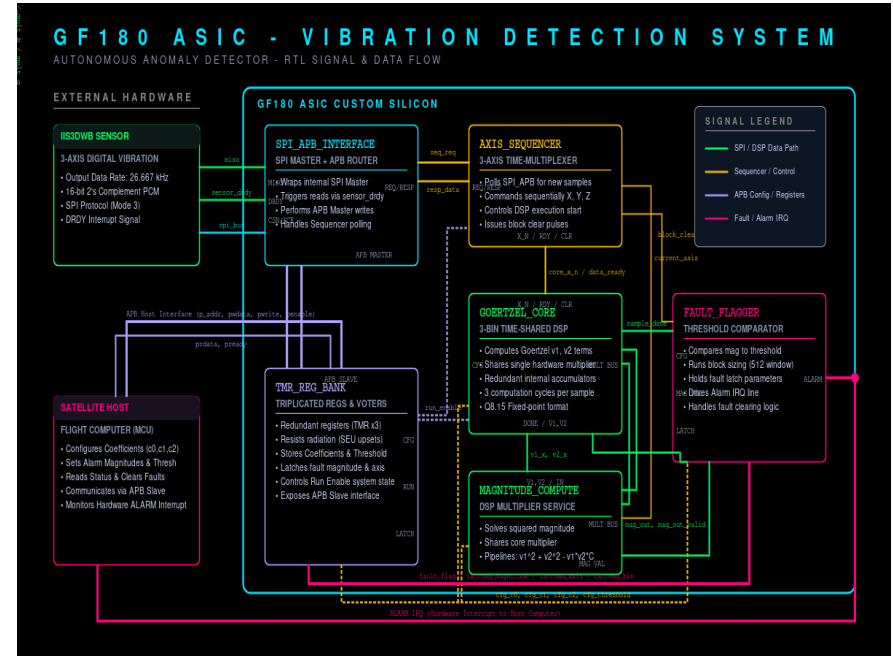
Direct Digital Sensor Interface (SPI & APB Master)

SPI Master Interface (Module 1)

- Protocol: SPI Mode-3 (CPOL=1, CPHA=1)
- Acquisition: 48-bit XYZ burst reads minimize transaction overhead
- Clock Domain Crossing: 2FF synchronizers prevent metastability
- Triggering: Event-driven by sensor DRDY interrupt
- Shared FSM: Seamlessly handles boot configuration (4 regs) & acquisition

APB Master & Bus Interface (Modules 2 & 3)

- Protocol: Standard AMBA APB protocol
- FSM Control: Lightweight 3-state control (IDLE, SETUP, ACCESS)
- Wait States: Full PREADY support allows slow peripherals
- Efficiency: Request-driven transactions minimize switching power



Hardware-Optimized DSP Engine (Goertzel Core)

The Goertzel Algorithm (Second-Order IIR Filter)

$$v[n] = x[n] + C \cdot v[n-1] - v[n-2]$$

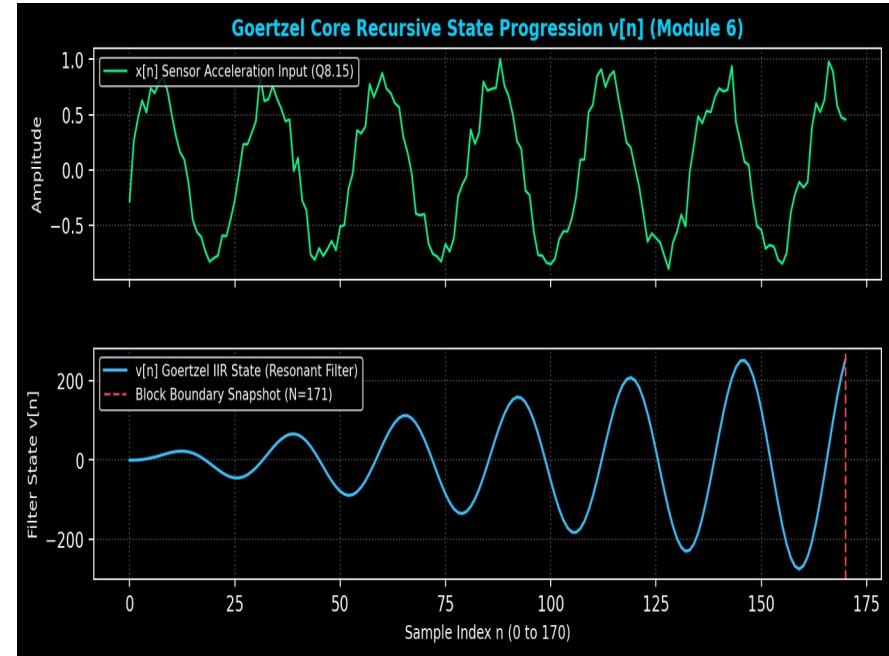
Why Goertzel over FFT?

Computes energy at targeted frequency bins (1k, 5k, 10k Hz) only.

- Bins monitored: 3 programmable bins
- Execution speed: 6 clock cycles per sample
- Control: Triplicated (TMR) state machine for SEU resilience

DSP Hardware Features (Module 6)

- Shared Multiplier: Single multiplier shared across 3 bins
- Saturating Q8.15 Fixed-Point: Prevents overflow corruption
- Operand Isolation: Inputs gated when core is idle
- Power Savings: Active only 6 cycles out of 1875 cycles per sample



$$\text{Magnitude}^2 = v1^2 + v2^2 - C \cdot v1 \cdot v2$$

(v1 = v[N-1], v2 = v[N-2] at block boundary)

At block boundary (N=171), captures v1, v2, coefficient C, and axis index.

- Multiplier Reuse: Magnitude engine reuses Goertzel core multiplier — 0 area cost!

- Triggering: Remains idle (near-zero power) until block boundary event

- **Axis Attribution:** Physical axis index snapshotted alongside Goertzel states



Radiation Hardening by Design (RHBD) Strategy

Space Radiation Protection

High-energy space particles flip register bits (Single Event Upsets / SEUs).

Triple Modular Redundancy (TMR)

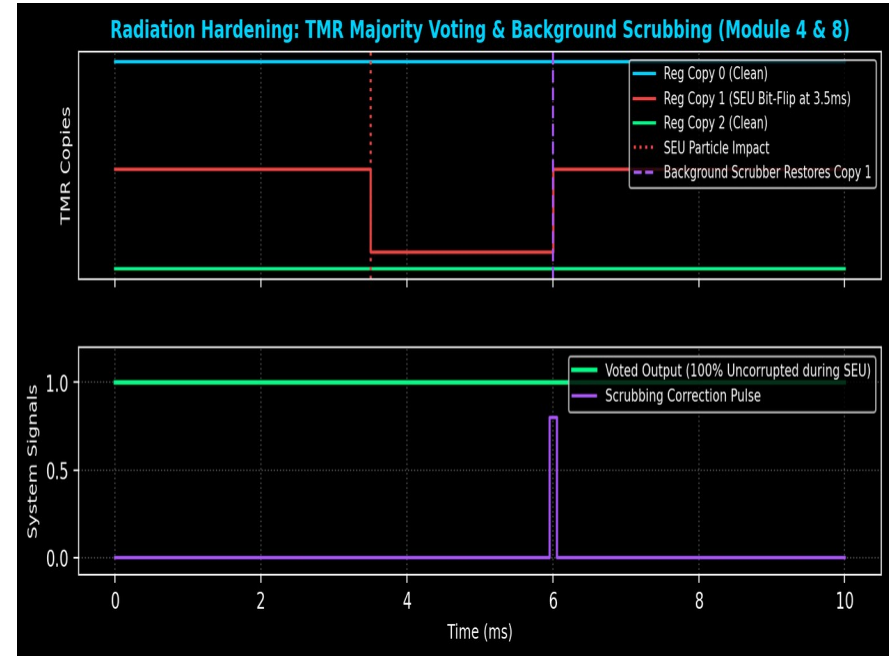
- Triplicated configuration registers, Goertzel coefficients, thresholds & FSMs.
- 1-cycle majority voting continuously resolves and masks bit flips.

Background Scrubbing (Module 4)

- Periodic scrubber checks register copies in background and restores corrupted copies.

SRAM-Free & Physical Hardening

- Hardened flip-flop arrays (no radiation-vulnerable SRAM).
- Dense substrate taps & guard rings enclose active logic.



Critical Area Constraint & Trade-off Options

Engineering Challenge: IIS3DWB outputs 3 axes simultaneously, but Chipathon die area is strictly capped at $600 \times 600 \mu\text{m}$.

Option	Architecture Description	Die Area Cost	Fault Detection Latency
Option 1	3 Parallel Goertzel Cores (X, Y, Z)	Exceeds $600 \times 600 \mu\text{m}$ limit ❌	Fastest (19.2 ms)
Option 2	Single-Axis Processing Only	Within Area Limit ✔️	No multi-axis coverage ❌
Option 3	Axis Sequencer (BLOCK_SIZE=171)	Fits $600 \times 600 \mu\text{m}$ Budget (0 Area Penalty) ✔️	3× Latency Speedup (~6.4 ms) ✔️

Selected Optimization: Option 3 (BLOCK_SIZE = 171) — Reduces block size from 512 to 171, achieving 3× faster per-axis latency detection with zero area penalty.

Option 3 Implementation & Trade-off Analysis

Quantitative Performance Trade-offs

- BLOCK_SIZE reduced from 512 → 171 (3× improvement in per-axis detection latency)
- Threshold scaling: Magnitude $\propto N^2$, so thresholds divided by $(512/171)^2 \approx 8.96$
- Area: Zero additional area cost — achieved through sequencing, not added hardware

Accepted Limitations & Spacecraft Justification

1. Sequential Processing Data Loss: 2 of 3 axes discarded during any single block acquisition.
2. Intermittent Fault Coverage: Cannot detect simultaneous multi-axis faults during rotation.
3. Justification for Spacecraft Applications:
Mechanical failures (bearing wear, pump fatigue) degrade slowly over hours or days.
Per-axis latency of ~6.4 ms is far below the timescale of structural degradation.
This trade-off is fully acceptable for the mission profile.

Comprehensive Verification & Regression Testing

100% Assertion Success — All 79 Verification Checks Passed

Top Integration (tb_top.v) — 7 / 7 Passed ✓

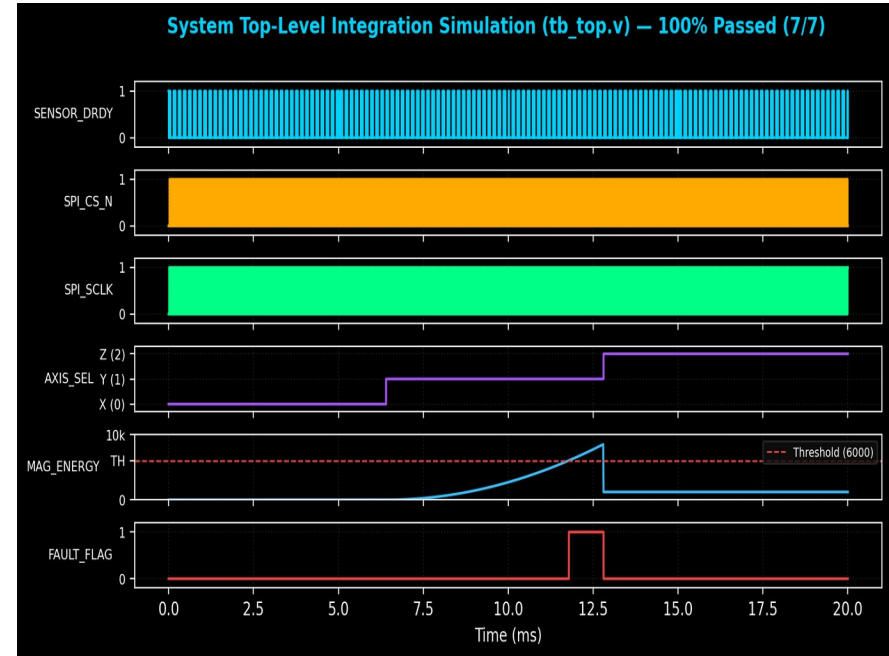
- Case 1: Quiet normal operation (no false positives)
- Cases 2-4: X, Y, Z-axis fault detected & flagged
- Timing: Confirmed 6.4 ms per-axis rotation intervals

Goertzel Core (tb_goertzel_core.v) — Passed ✓

- Verified second-order recurrence filtering
- Q8.15 fixed-point arithmetic vs. Python model

SPI Master Interface — 71 / 71 Passed ✓

- Boot config, DRDY sync, Mode-3 timing



Roadmap & Key Takeaways

Key Contributions

1. Direct Digital MEMS Interface: Eliminates analog radiation-vulnerable ADCs.
2. Pragmatic Optimization (Option 3): Fits $600 \times 600 \mu\text{m}$, $3\times$ faster detection, zero area cost.
3. Radiation Hardening (RHBD): TMR, majority voting, background scrubbing.
4. Verified & Tapeout-Ready: 79/79 verification tests passed.

Project Roadmap

- Q3 2026: Tape-out via SSCS Chipathon Shuttle (GF180MCU process)
- Q4 2026: Post-silicon testing and power/latency measurement
- Q1 2027: Implement Option 1 (Parallel Cores) — eliminate sequential data loss
- Q2 2027: Flight qualify for CubeSat missions (TRL 5/6); open-source RTL release